Lab 07: Moderated Models.

Question 1. "Simple" versus Interactive Effects

In our multivariable regression output from R, we get the following when we regress the DECREASE in a person's VO2max onto the ALTITUDE at which the test was conducted (in km), that person's baseline VO2max (BVO2, in mL/kg/min), the interaction of those two terms:

```
Coefficients:
            Estimate Std. Error t value Pr(>|t|)
(Intercept)  8.51588    3.01436   2.825 0.005697 **
alt_km      -0.11323    1.00029  -0.113 0.910097
BVO2        -0.10874    0.05266  -2.065 0.041512 *
alt_km:BVO2 -0.07448    0.01849  -4.027 0.000109 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.432 on 101 degrees of freedom
Multiple R-squared:  0.8488,     Adjusted R-squared:  0.8443
F-statistic:   189 on 3 and 101 DF,  p-value: < 2.2e-16
```

Extracting the parameter estimates, we get the following equation:

$$\widehat{DECREASE}_i = 8.516 - 0.113(ALT_i) - 0.109(BVO2_i) - 0.074(ALT * BVO2_i)$$

Given this equation, reduce it to simpler equations to show:

- The effect of altitude for someone with a BVO2 = 33 mL/kg/min:
  - $[8.516 - 0.109(33)] + [-0.113 - 0.074(33)](ALT)$
  - $4.919 - 2.555(ALT_i)$

- The effect of altitude for someone with a BVO2 = 64 mL/kg/min:
  - $[8.516 - 0.109(64)] + [-0.113 - 0.074(64)](ALT)$
  - $1.540 - 4.849(ALT_i)$

- The effect of BVO2 for someone at sea-level, ALT = 0 Km:

$$\widehat{DECREASE}_i = [8.516 - 0.113(0)] + [-0.109 - 0.074(0)](BVO2_i)$$

$$\widehat{DECREASE}_i = 8.516 - 0.109(BVO2_i)$$

- The effect of BVO2 for someone at sea-level, ALT = 3 Km:

$$\widehat{DECREASE}_i = [8.516 - 0.113(3)] + [-0.109 - 0.074(3)](BVO2_i)$$

$$\widehat{DECREASE}_i = 8177 - 0.331(BVO2_i)$$

Question 2. Changing what X = 0 means.

Next, we will analyze the same dataset but we will use a "mean-centered" version of the BVO2 variable. That is:

$$BVO2.c_i = BVO2_i - \overline{BVO2} = BVO2.c - 53.525$$

Not that ALTITUDE is the elevation at which the test was conducted (in km), but that BVO2.c is now the difference between a person's BVO2 and the average baseline VO2 (53.525 mL/kg/min). Including these factors and their interaction in a model predicting the DECREASE in VO2max at altitude, we get:

```
Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    2.69580    0.59754   4.512 1.74e-05 ***
alt_km        -4.09992    0.17710 -23.150  < 2e-16 ***
BVO2.c        -0.10874    0.05266  -2.065 0.041512 *
alt_km:BVO2.c -0.07448    0.01849  -4.027 0.000109 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 2.432 on 101 degrees of freedom
Multiple R-squared:  0.8488,     Adjusted R-squared:  0.8443
F-statistic:   189 on 3 and 101 DF,  p-value: < 2.2e-16
```

Extracting the parameter estimates, we get the following equation:

$$\widehat{DECREASE}_i = 2.696 - 4.100(ALT_i) - 0.109(BVO2.c_i) - 0.074(ALT * BVO2.c_i)$$

The first thing to notice is that a few things in our output have changed.

- Why has the value of the intercept changed in this model (with BVO2.c) compared to the previous model (with BVO2)?

- Notice that the parameter for "alt_km" is now statistically significant ($p < 0.001$) when it wasn't before. Why has the effect of ALTITUDE changed in this model (with BVO2.c) compared to the previous model (with BVO2)?

Given this equation, reduce it to simpler equations to show:

- The effect of ALTITUDE for someone with a BVO2.c = +4.5 mL/kg/min:

$$\widehat{DECREASE}_i = 2.696 - 4.100(ALT_i) - 0.109(BVO2.c_i) - 0.074(ALT * BVO2.c_i)$$

$$[2.696 - 0.109(4.5)] + [-4.100 - 0.074(4.5)](ALT_i)$$

$$2.206 - 4.433(ALT_i)$$

- The effect of ALTITUDE for someone with a BVO2.c = -10 mL/kg/min:

$$[2.696 - 0.109(-10)] + [-4.100 - 0.074(-10)](ALT_i)$$

$$3.786 - 3.360(ALT_i)$$

- The effect of ALTITUDE for someone with an average BVO2:

$$[2.696 - 0.109(0)] + [-4.100 - 0.074(0)](ALT_i)$$

$$2.696 - 4.100(ALT_i)$$

- Find the predicted decrease in VO2max for someone with a baseline VO2 of 63 when test at an elevation of 5km:

Mean Centered Equation:

$$\widehat{DECREASE}_i = 2.696 - 4.100(5) - 0.109(63 - 53.525) - 0.074(5 * (63 - 53.525))$$

$$\widehat{DECREASE}_i = -22.343$$